2.3.4 Hamiltonian Flow Maps are Symplectic

As we saw at the end of the last chapter, the variational equations describe the change in an infinitesimal perturbation of a solution of a dynamical system. For the Hamiltonian system $\dot{z} = \boldsymbol{J}\nabla H(z)$, these take the form:

$$\dot{\boldsymbol{W}} = \boldsymbol{J}\boldsymbol{S}(t)\boldsymbol{W},$$

where $\boldsymbol{S}(t) = H_{zz}(\boldsymbol{z}(t, \boldsymbol{\zeta}))$ is a symmetric matrix. Computing

$$\boldsymbol{W}^T\boldsymbol{J}\dot{\boldsymbol{W}} = \boldsymbol{W}^T\boldsymbol{J}^2\boldsymbol{S}\boldsymbol{W} = -\boldsymbol{W}^T\boldsymbol{S}\boldsymbol{W},$$

whereas

$$\dot{\boldsymbol{W}}^T\boldsymbol{J}\boldsymbol{W} = \boldsymbol{W}^T\boldsymbol{S}^T\boldsymbol{J}^T\boldsymbol{J}\boldsymbol{W} = \boldsymbol{W}^T\boldsymbol{S}\boldsymbol{W},$$

hence

$$\frac{d}{dt}\boldsymbol{W}^T\boldsymbol{J}\boldsymbol{W} = \boldsymbol{W}^T\boldsymbol{J}\dot{\boldsymbol{W}} + \dot{\boldsymbol{W}}^T\boldsymbol{J}\boldsymbol{W} = 0.$$

This means that $\boldsymbol{W}^T\boldsymbol{J}\boldsymbol{W}$ is a constant matrix. Observe that $\boldsymbol{W}(t) = \mathscr{F}_t'(\boldsymbol{z}(t, \boldsymbol{\zeta}))$ and that $\boldsymbol{W}(0) = \mathscr{F}_t'(\boldsymbol{z}(t, \boldsymbol{\zeta}))\big|_{t=0} = I$, hence

$$\boldsymbol{W}^T\boldsymbol{J}\boldsymbol{W} \equiv \boldsymbol{W}(0)^T\boldsymbol{J}\boldsymbol{W}(0) = \boldsymbol{J}.$$

This proves that the flow map of a Hamiltonian system is a symplectic map.

2.3.5 The Symplectic Maps Form a Group

Let $\boldsymbol{\Phi}_1$ and $\boldsymbol{\Phi}_2$ be any pair of symplectic maps. Then

$$(\boldsymbol{\Phi}_1 \circ \boldsymbol{\Phi}_2)' = \boldsymbol{\Phi}_1'\boldsymbol{\Phi}_2',$$

by the chain rule, hence

$$[(\boldsymbol{\Phi}_1 \circ \boldsymbol{\Phi}_2)']^T\boldsymbol{J}(\boldsymbol{\Phi}_1 \circ \boldsymbol{\Phi}_2)' = [\boldsymbol{\Phi}_1'\boldsymbol{\Phi}_2']^T\boldsymbol{J}\boldsymbol{\Phi}_1'\boldsymbol{\Phi}_2' = \boldsymbol{\Phi}_2'^T\boldsymbol{\Phi}_1'^T\boldsymbol{J}\boldsymbol{\Phi}_1'\boldsymbol{\Phi}_2' = \boldsymbol{\Phi}_2'^T\boldsymbol{J}\boldsymbol{\Phi}_2' = \boldsymbol{J}.$$

Thus the composition of any pair of symplectic maps is a symplectic map. The determinant of a symplectic map is ± 1, hence these maps are always invertible, and the inverse of a symplectic map is symplectic since $\boldsymbol{\Phi}'^T\boldsymbol{J}\boldsymbol{\Phi}' = \boldsymbol{J}$ implies $\boldsymbol{J} = \boldsymbol{\Phi}'^{-T}\boldsymbol{J}\boldsymbol{\Phi}'^{-1}$. Thus the symplectic maps form a group under composition.